MiniMax

1.

```
def minimax(tree, values, node, is_max):
    if node not in tree:
        return values[node]
    result = []
    for child in tree[node]:
        result.append(minimax(tree, values, child,
not is_max))
    if is_max:
        return max(result)
    else:
        return min(result)
tree = {}
values = {}
n = int(input("Enter number of non-terminal nodes:
"))
for i in range(n):
    node = input("Enter node name: ")
    children = input("Enter child nodes separated
by space: ").split()
    tree[node] = children
leaves = []
for children in tree.values():
    for child in children:
        if child not in tree and child not in
leaves:
            leaves.append(child)
print("Terminal nodes:", " ".join(leaves))
for leaf in leaves:
    values[leaf] = int(input("Enter value of " +
leaf + ": "))
root = input("Enter root node: ")
answer = minimax(tree, values, root, True)
print("Minimax value:", answer)
```

```
Enter child nodes separated by space: B C
Enter node name: B
Enter child nodes separated by space: D E
Enter node name: C
Enter child nodes separated by space: F G
Enter node name: D
Enter child nodes separated by space: 2 3
Enter node name: E
Enter child nodes separated by space: 5 9
Enter node name: F
Enter child nodes separated by space: 0 1
Enter node name: G
Enter child nodes separated by space: 7 5
Terminal nodes: 2 3 5 9 0 1 7
Enter value of 2: 2
Enter value of 3: 3
Enter value of 5: 5
Enter value of 9: 9
Enter value of 0: 0
Enter value of 1: 1
Enter value of 7: 7
Enter root node: A
Minimax value: 3
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Visualize Ru